Juan de Dios Alvarez

Data Analyst -- Commercial and Operations Reporting, KPIs and Statistical Modeling

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Profile

Data analyst and data scientist with seven years of industry work (2019 to present), four of them with US-based clients in a distributed, English-speaking team. I consolidate data from operational systems, supplier files and public sources into reports, indicators and dashboards that business teams use to decide. My work covers the full path: understanding what the business needs, extraction through APIs and SQL, data preparation, analysis in Python and R, and presentation in Power BI, Excel or Tableau, with each figure traceable to its source. I look for manual steps that can be automated and propose the improvement.

Selected Results

- **Point-of-sale and inventory data:** integrated point-of-sale, supplier, e-invoice and bank data for a small retail business in Mexico into one PostgreSQL warehouse, with FIFO inventory valuation and monthly Excel sales and stock reports.
- **Indicators for decision-makers:** at FIRA, a development bank, automated the data behind credit-assignment and non-payment-risk analysis: 74 scheduled extraction scripts over 9+ public sources, 60+ SQL functions, automated executive Excel reports and 15+ dashboards.
- **Power BI for commercial teams:** dashboard that told a US client's sales team which products to offer each school district; executive and operational dashboards for non-technical decision-makers.
- **Process improvement:** replaced an analyst's manual Excel process with a scheduled pipeline any team member can run.

Professional Experience

Data Scientist & ML Engineer | Independent Practice

March 2026 -- present

- Designed and run in production the data system of a small retail business in Mexico: PostgreSQL warehouse (40+ tables) integrating point-of-sale, supplier, e-invoice and bank data, used daily by store staff.
- Reconciled product identity across supplier catalogs, invoices and the point-of-sale system, which share no common key, and integrated two supplier product and price APIs into one catalog.
- Built FIFO cost-layer inventory valuation and defined stock value, on-hand quantity and trailing sales once, so every report and screen gives the same figures.
- Designing automatic product pricing (in progress): rule-based engine that sets prices from supplier cost, exchange rate, latest invoice cost, markups by customer type and tax, with a log of price changes.
- Produced scheduled Excel reports on monthly sales and inventory value, each with notes on method and data coverage so business users can read them on their own.
- Audited every supplier invoice line against what reached the warehouse, sorting each gap by root cause.

Economic Research Specialist (Data Scientist) | FIRA

January 2025 -- January 2026

- Replaced one analyst's manual Excel process, which fed credit-assignment and non-payment-risk analysis, with a scheduled, unattended pipeline: 74 extraction scripts (Python, API connections, Playwright) across 9+ public sources (INEGI, Banxico, SNIIM, SIAP, USDA) loading PostgreSQL with provenance metadata.
- Wrote 60+ PostgreSQL functions and materialized views computing indicators: inflation contributions, trade balances, supply-and-demand balances and crop yields.
- Delivered automated executive Excel reports and 15+ Apache Superset dashboards to decision-makers; set governance conventions (raw data preserved, lineage, GitLab) and supervised an undergraduate team.

- Built a crop-production estimation pipeline from Sentinel-2 vegetation-index time series for 310 plots (pandas, scikit-learn): Random Forest and Gradient Boosting regressors compared by 5-fold cross-validation, trained on a proxy target because field-level yields were not available.

Data Scientist (contract) | CITEIM

October 2024 -- January 2025

- Built the dataset for a Mexican Sign Language recognition system from video (OCR, speech-to-text, MediaPipe keypoints).

Data Scientist & Team Lead | Punto Singular

December 2023 -- August 2024

- Led monthly analytics cycles for an EdTech client, from problem and metric definition to results presented against the agreed metric; KPI and quality dashboards in R, Python, Tableau and Looker used directly by business stakeholders; Power BI and data-flow design for LMS analytics.

Data Scientist (contract) | BairesDev

September 2023 -- November 2023

- Built a retrieval-augmented generation proof of concept for querying a legal client's contracts for obligations and risk clauses.

Data Scientist | D3Clarity (formerly Viqtor Davis North America)

February 2019 -- March 2023

- First data scientist of the company; defined the analytics roadmap and delivered it to US-based clients in a distributed team.
- Built executive and operational dashboards in Power BI and Tableau for non-technical decision-makers; extracted data from a master-data system into Power BI reports.
- Built forecasting models to identify consumer segments for client sales and marketing strategy; data lakes, ETL and master data management (Semarchy xDM).
- Classified public school-board meeting minutes with NLP on Azure ML, from proof of concept to production, feeding a Power BI dashboard for the client's sales team.

Earlier Experience

- Lecturer, Tecnologico de Monterrey Business School (KPI dashboarding in Tableau for economics and management students); HAWC observatory data analysis; data-acquisition software at CERN.

Education

- **Master's in Computer Science** -- INAOE, Mexico. Degree awarded; SEP professional license (cedula) 5916605.
- **PhD Candidate in Physics** -- University of Michoacan. Computational research: large-scale simulations of cosmic-ray air showers; principal author of two ICRC 2017 papers; co-author on 14 publications.
- **Bachelor's in Physics and Mathematics** -- University of Michoacan.
- **Certifications:** AWS Certified Cloud Practitioner; Semarchy xDM. **Languages:** Spanish (native), English (professional), Italian (basic-intermediate), French (basic).

Technical Skills

- **Reporting and visualization:** Power BI, Excel (pivot tables, Power Query, automated reports), Tableau, Apache Superset, Looker
- **Data and integration:** SQL, PostgreSQL (PL/pgSQL, window functions, materialized views), MySQL, Firebird; REST API extraction, ETL, data lakes, master data management
- **Analysis and modeling:** Python (pandas, NumPy, scikit-learn), R, Random Forest, Gradient Boosting, PCA, forecasting
- **Cloud and business systems:** Azure Machine Learning, AWS; SAP (end user)